



ElectronicsTutorials

RESISTIVE CIRCUITS

For Students, Professionals
and Beyond

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1. RESISTOR AND RESISTANCE

Electrical Resistance is a measure of how much a component opposes or resists the flow of electrical charge around a closed circuit. The amount of electrical charge that can flow in a circuit depends upon the magnitude of the voltage provided by the voltage source.

However, electrical charge in the form of a current also depends on the resistance that the circuit offers to this flow of charge as electrical resistance defines the relationship between voltage and current flow. The basic circuit element which demonstrates this current-voltage (I-V) relationship is called a **Resistor**.

Resistors are the most fundamental and commonly used of all electronic components, to the point where they are almost taken for granted, but they play a vital role within any electrical or electronic circuit.

The principal job of a resistor within a circuit is to “resist” (hence the name Resistor), regulate or set a constant flow of electrons (current) through them and as such are commonly used as voltage dividers or current limiters within a circuit.

Resistors add electrical resistance to a circuit so by connecting them together in different series or parallel combinations, any value of circuit resistance can be obtained.

Resistors are passive components which resist the flow of electrical current around a closed circuit. The unit of resistance is the Ohm

Resistors are what are called “Passive Electrical Components”, that is they contain no power source or provide any form of amplification. Instead resistors reduce or attenuate the voltage or current signal passing through them. This attenuation results in electrical energy being lost or dissipated in the form of heat as the resistor tries to resist the flow of electrons through itself.

The SI unit of resistance is the Ohm. To identify resistance it is given the Greek symbol Ω (Omega). A conductor is said to have a resistance of one ohm when one volt causes one ampere of current to flow through it. Note that Resistance cannot be negative in value only positive and in AC circuits, AC resistance equals DC resistance, $Z_R = R$.

TABLE 1. PREFIXES USED FOR OHMS

Prefix for Ohms	Symbol	Description	Abbreviation	Value
Megaohm	M	One Million Ohms	$M\Omega$	$10^6 \Omega$
Kilohm	k	One Thousand Ohms	$k\Omega$	$10^3 \Omega$
Ohm	-	One Ohm	Ω	$10^0 \Omega$
Milliohm	m	One Thousandth of an Ohm	$m\Omega$	$10^{-3} \Omega$
Microhm	μ	One Millionth of an Ohm	$\mu\Omega$	$10^{-6} \Omega$

The most common prefixes used in Table 1 above to indicate the value of resistance are: Kilo-ohms ($k\Omega = 10^3\Omega$) and Mega-ohms ($M\Omega = 10^6\Omega$).

The amount or value of resistance a particular circuit element has determines whether the element is a “good conductor” with a low value resistance, a “bad conductor” (insulator) with high value resistance, or has a value somewhere in between. Figure 1 shows that an electrical cable consists of both an insulator and an electrical conductor.

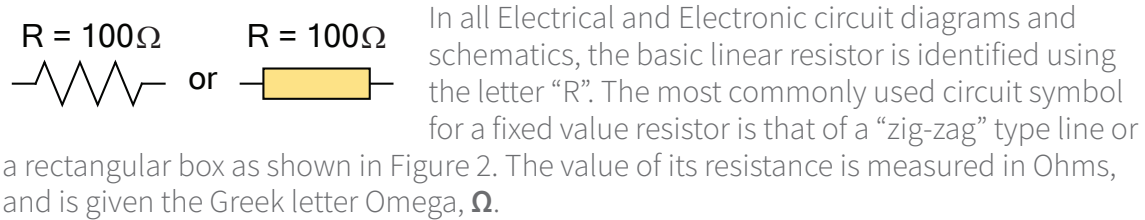
FIGURE 1. AN ELECTRICAL CABLE



Low resistance, for example one ohm or less implies that the circuit is a good conductor made from materials with lots of free electrons in its valence shell to pass a current.

Examples of good conductors are generally metals such as copper, aluminium, gold, silver or non-metals such as soft carbon, graphite or mercury.

High resistance, one mega-ohm or more implies the element is a bad conductor of electricity. Bad conductors in the form of insulators are generally made from non-metallic materials with no free electrons, or they have tightly bound electrons in their valence shell. Examples of non-conductive insulators include glass, porcelain, rubber, plastics and dry wood or paper, etc.

FIGURE 2. RESISTOR CIRCUIT SYMBOL


Resistors can have resistance values ranging from less than one ohm, ($<1\Omega$) to well over tens of millions of ohms, ($>10M\Omega$) in value.

The factors which affect any resistance of a uniform size are:

1. **Length of Material:** The resistance of a material is directly proportional to its length. The longer the length of the material the more resistance it has.
2. **Cross-sectional Area:** The resistance of a material is indirectly proportional to its width. The wider or thicker the material the less resistance it has allowing more free electrons to flow.
3. **Type of Material:** The resistivity of the type of material affects the amount of free electrons able to flow through it. A conductor has less resistance while an insulator has more resistance.
4. **Temperature:** The temperature of the material affects its resistance. The higher the temperature, the higher the resistance. Materials such as thermocouples and thermistors are design to change their resistance with temperature.

2. RESISTIVITY

The type of material used to construct a resistor, along with its physical size and shape all determine the value of its resistance and we can create a resistor by simply attaching connections to any type of conductive material.

However, because every material has different properties and will therefore oppose the flow of current in different ways, a materials electrical resistance is determined by the properties of the actual material being used as well as its physical dimensions.

Resistivity is a measure of how good or bad a material is at conducting an electrical current

The specific electrical resistance of a material is known as its resistivity, ρ (Greek letter Rho) and is a measure of how strongly the material will oppose the flow of electric current through itself. Electrical Resistivity (ρ) is measured in units of ohm meters, (Ωm) and is given as:

$$\rho = \frac{R \times A}{L} \Omega m$$

Where: ρ is electrical resistivity, R is the resistance in ohms (Ω), L is the length in metres (m), A is the area in square metres (m^2)

This resistivity factor allows different types of materials to be compared to one another at a specified temperature according to their physical properties without regards to their lengths or cross-sectional area. Thus a particular material can be classified as being either a conductor, an insulator or a semiconductor material depending upon the value of their resistivity and the higher the resistivity value of ρ , the more resistance and vice versa.

For example, the resistivity of a good conductor such as copper is in the order of 1.72×10^{-8} ohm metre (or $17.2 \text{ n}\Omega m$), whereas the resistivity of an insulator, such as air can be well over 1.5×10^{14} or 150 trillion Ωm . Obviously zero resistivity would mean zero resistance. Superconductors can have zero resistance at low temperatures.

Materials such as copper and aluminium are known for their low levels of resistivity thus allowing electrical current to easily flow through them. This makes these materials ideal for use in electrical conductors, wires and cables. Silver and gold do have much low resistivity values, but for obvious reasons are more expensive to turn into electrical wires.

While the resistivity of a material may decide whether the material is a good conductor allowing the generous flow of electrical charge, or a good insulator (poor conductor). By altering the chemical composition of the base material from which a resistor is made, it is possible to create resistances of any resistive value for a given physical size.

Resistors are generally made using a carbon (graphite) conductor for general purpose carbon resistors (CR) or with a semiconductor substrate for metal film (MF) or metal oxide precision resistors.

By altering the ratio of doping concentration to base conductor material, it is possible to change the conductivity and therefore the resistance value to create a resistor of a fixed known value. The resistor body is colour coded to identify its resistive value.

3. TYPES OF RESISTORS

Resistors are readily available in many types, shapes and wattage sizes from small surface mount chip resistors to large wire wound heat dissipating resistors. However, resistors can be classified into the following three basic types.

- **Fixed Value Resistors** - A fixed value resistor has one single value of resistance, for example 100Ω . The actual value of its resistance is determined by its percentage tolerance range either side of its nominal value. For circuit analysis methods, its resistance does not change producing a straight line linear response.
- **Variable Resistors** - A variable resistor such as a potentiometer or rheostat can produce either a linear or logarithmic response. It can provide an infinite number of resistive values between zero and its maximum resistive value. For example 0Ω to $1,000\Omega$. This is achieved by rotating a shaft or adjusting a slider to move a wiper.
- **Non-Linear Resistors** - Non-linear resistors are resistive devices whose resistive value changes significantly in response to an externally applied stimulus. Such as temperature, light, pressure or voltage. Non-linear resistors include light dependent resistors, or LDR's, thermistors, photo-resistors, strain gauge's and varistor's.

FIGURE 3. RESISTOR SYMBOLS

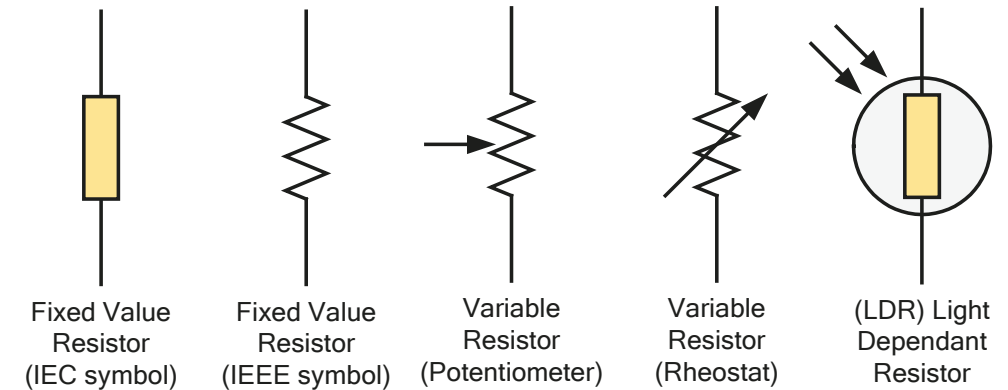


Figure 3 shows the different symbols used for the three main types of resistors. The two most common types of fixed ohmic value resistors are *wirewound* and *carbon composition*. Carbon composition resistors are generally used when large resistances are needed.

While wirewound resistors, with their metal finned or epoxy encased body are used for very high wattage applications. Variable resistors, known as potentiometers or rheostats are also available.

4. FIXED VALUE RESISTORS

Fixed resistors are manufactured with a specified and fixed value of resistance. Fixed value resistors are the most common and easily recognizable of all the passive components. These resistors have only two electrical connections allowing electrical current to flow in either direction through them.

Thus fixed value resistors are bi-directional devices with no specific polarity offering the same amount of resistive value regardless of the direction in which the current flows.

Fixed value resistors are available as through-hole components with external leads for insertion and soldering onto a circuit board (PCB) or as chip resistors for surface mount technology (SMT) requiring physically small size resistors that are soldered directly onto the surface of a circuit board.

All modern fixed value resistors can be classified into four broad groups:

- **Carbon Composition Resistor** – Made from compressed carbon dust or graphite paste, available in low wattage values
- **Film or Cermet Resistor** – Made from a conductive metal oxide paste, very low wattage values
- **Wire-wound Resistor** – Metallic or ceramic bodies for heatsink mounting, used for very high wattage power ratings
- **Semiconductor Resistor** – These are high frequency, precision made surface mount thin film technology

There are a large variety of fixed and variable resistor types with different construction styles available for each group, with each one having its own particular characteristics, advantages and disadvantages compared to the others.

The principal parameters for a resistor, whether it is fixed or variable type are:

- **Nominal Resistance Value** expressed in Ohms (Ω)
- **Power Rating** given in Watts (W)
- **Resistance Tolerance** expressed as a percentage of its fixed value, e.g. $\pm 10\%$
- **Temperature Coefficient** in parts per million (ppm) by which the fixed value changes with temperature

The two most obvious characteristics of any fixed resistor are its nominal resistance value and its maximum power rating.

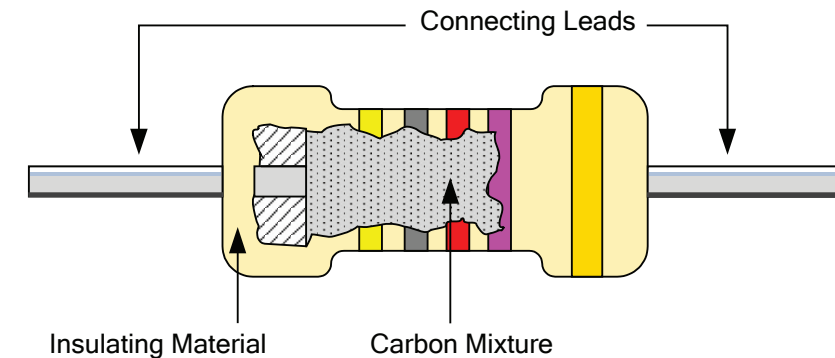
4.1 Carbon Composite Resistors

Carbon Composition Resistors (CCR) are constructed from a solid rod of conductive composite material with the most common type being the *Carbon Resistor*. Carbon resistors are a cheap general purpose resistor used in electrical and electronic circuits.

Their resistive element is manufactured from a mixture of finely ground carbon dust or graphite (similar to pencil lead) and a non-conducting ceramic (clay) powder to bind it all together before being protected with a special epoxy coating and then colour code marked for identification.

The ratio of carbon dust to ceramic (conductor to insulator) determines the final resistive value of the mixture and the higher the ratio of carbon to filler, the lower the resistance. Carbon resistors can have resistance values ranging from one ohm, (1Ω) to well over tens of millions of ohms, ($10M\Omega$) in resistive value.

FIGURE 4. CARBON RESISTOR



Carbon Resistors shown in Figure 4, are a low to medium type power resistor having a low inductance value, making them ideal for use in high frequency applications, but they can also suffer from noise and instability when hot. Carbon composite resistor types are very cheap to make and are therefore commonly used in electrical circuits.

Carbon resistors are usually prefixed with a “CR” notation, for example CR10k Ω . They are available in E6 ($\pm 20\%$ tolerance (accuracy)), E12 ($\pm 10\%$ tolerance) and E24 ($\pm 5\%$ tolerance) packages with power ratings from 0.250 or 1/4 of a Watt up to 5 Watts. However, due to their manufacturing process carbon type resistors tend to have very large tolerances.

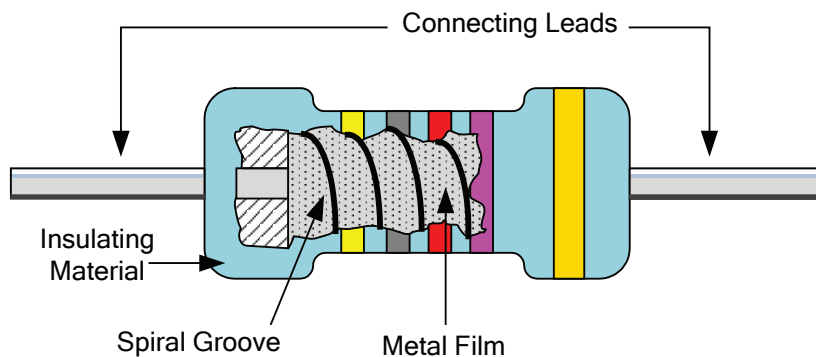
4.2 Film Type Resistors

Film Resistors is the term used for Metal Film, Carbon Film and Metal Oxide Film types, which are made by depositing pure metals, such as nickel, or an oxide film, such as tin-oxide, onto an insulating ceramic rod or base substrate.

Their resistive value is precisely controlled by increasing the desired thickness of a conductive film deposited onto the substrate giving them the more familiar name of either “thick-film resistors” or “thin-film resistors”.

Once the conductive film is deposited, a laser is used to cut a high precision spiral helix groove type pattern into this film. The cutting of the film has the effect of increasing the conductive or resistive path, a bit like taking a long length of straight wire and forming it into a coil.

FIGURE 5. METAL FILM RESISTOR



This method of manufacture allows for much closer tolerance resistors (1% or less) as compared to the simpler carbon composition types producing smaller sized resistors with much higher ohmic values, in excess of 10M Ω (10 Million Ohms) compared to other types.

Metal Film Resistors shown in Figure 5, have much better temperature stability than their carbon equivalents, lower noise and are generally better for high frequency or radio frequency applications.

Fixed value resistors can be available as carbon, metal film, metal oxide thick film or wirewound types

Therefore a much better choice when any degree of precision or stability is required. **Metal Oxide Resistors** have better high surge current capability with a much higher temperature rating than the equivalent metal film resistors.

Another type of metal film resistor commonly known as a **Thick Film Resistor** is manufactured by depositing a much thicker conductive paste of **CER**amic and **MET**al, called *Cermet*, onto an alumina ceramic substrate.

Cermet resistors have similar properties to metal film resistors and are generally used for making small surface mount chip type resistors, multi-resistor networks in one package for PCB and high frequency resistors. They have good temperature stability, low noise, and good voltage ratings but low surge current properties.

The abbreviations and markings can sometimes give a clue to the type of resistor. Metal Film Resistors are commonly prefixed with an “MFR” notation. For example MFR100k Ω . While a CF is used for Carbon Film types.

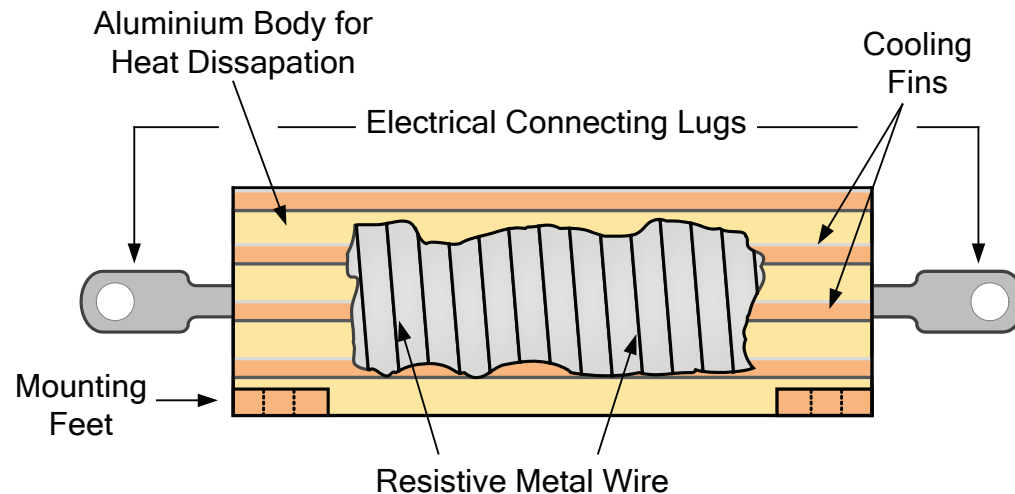
Metal film resistors are available in E24 ($\pm 5\%$ & $\pm 2\%$ tolerances), E96 ($\pm 1\%$ tolerance) and E192 ($\pm 0.5\%$, $\pm 0.25\%$ & $\pm 0.1\%$ tolerances). Power ratings ranging from 0.05 (1/20th) of a Watt up to 1/2 Watt.

Generally speaking film resistors and especially metal film resistors are precision low power components and are available for most surface-mount applications. Thick-film metal-ceramic composites and thin-film types are also available having better electrical characteristics as compared to carbon film resistor.

4.3 Film Type Resistors

Wirewound Resistors shown in Figure 6, are made by winding a thin metal alloy wire (Nichrome) onto an insulating ceramic former in the form of a spiral helix in a similar design to the film resistor above.

FIGURE 6. WIREWOUND RESISTOR



Wirewound resistors are generally only available in very low ohmic but high precision values ranging from 0.01Ω to about $100k\Omega$ due to the gauge of the wire and number of possible turns wound onto the base former.

Wirewound resistors are available in many different shapes and styles and are able to handle much higher electrical currents than other resistors of the same ohmic value with power ratings in excess of 300 Watts available.

These high power resistors are moulded or pressed into an aluminium heat sink body with fins attached to increase their overall surface area to promote heat loss and cooling.

Chassis Mounted Wirewound Resistors are designed to be physically bolted or mounted onto heatsinks or metal plates to further dissipate the generated heat. The mounting of the resistor onto a heatsink increases their current carrying capabilities even further.

Another type of wirewound resistor is the **Power Wirewound Resistor**. These are high temperature, high power non-inductive resistor types generally coated with a vitreous or glass epoxy enamel for use in resistance banks or DC motor/servo control and dynamic braking applications. They can even be used as low wattage space or cabinet heaters.

The non-inductive resistance wire is wound around a ceramic or porcelain tube covered with mica to prevent the alloy wires from moving when hot. Wirewound resistors are available in a variety of resistance and power ratings.

One of the main uses for power wirewound resistors is the electrical heating elements of an electric fire which converts the electrical current flowing through it into heat energy, with each element dissipating up to 1000 Watts, (1kW) of electrical energy.

Wirewound resistor types are prefixed with a “WH” or “W” notation (eg WH10 Ω) and are available in the WH aluminium clad package ($\pm 1\%$, $\pm 2\%$, $\pm 5\%$ and $\pm 10\%$ tolerance) or the W vitreous enamelled package ($\pm 1\%$, $\pm 2\%$ and $\pm 5\%$ tolerance) with power ratings from 1W to 300W or more.

5. PREFERRED RESISTOR VALUE AND TOLERANCES

Fixed value resistors come in a variety of sizes and values but to have a resistor available of every possible resistance value available, literally hundreds of thousands, if not millions of individual resistors would need to exist. Instead, resistors are manufactured in what are commonly known commonly as “preferred values”.

Instead of sequential values of resistance from 1Ω and upwards, certain values of resistors exist within certain tolerance limits. The tolerance of a resistor is the maximum difference it can vary between its actual real value and the stated value for it to still be acceptable.

Resistor tolerance is generally expressed as a plus or minus percentage value. For example, a $1\text{k}\Omega \pm 20\%$ tolerance resistor may have a maximum and minimum resistive value of:

Maximum Resistance Value of: $1\text{k}\Omega$ or $1000\Omega + 20\% = 1,200\Omega$

Minimum Resistance Value of: $1\text{k}\Omega$ or $1000\Omega - 20\% = 800\Omega$

Thus a $1\text{k}\Omega \pm 20\%$ tolerance resistor can have a maximum value of **1200 Ω** and a minimum value of **800 Ω** . This results in a difference in resistance of 400 Ω (1200 - 800) even for the same value resistor, or between different resistors of the same ohmic value.

E-Series resistance values as shown in Table 2, where created to standardise the resistors available while still allowing for a broad range of resistances to cover most needs. The tolerances for standard fixed value resistors are typically $\pm 20\%$, $\pm 10\%$, $\pm 5\%$ or $\pm 1\%$.

Tolerances as tight as 0.5%, 0.25%, and even 0.1% or higher are available but obviously the resistors with the smallest tolerance are the most accurate and the most expensive.

TABLE 2. RESISTOR TOLERANCE AND E-SERIES TABLE

Series	Resistor Tolerance	Values per Decade	Usage
E6	20%	6 steps per decade	No longer in use
E12	10%	12 steps per decade	Not commonly used
E24	5%	24 steps per decade	Commonly used
E48	2%	48 steps per decade	Commonly used
E96	1%	96 steps per decade	Commonly used
E192	0.5%, 0.25%, 0.1%	192 steps per decade	Precision Usage

The mathematical basis behind these preferred values comes from the square root value of the actual series being used. For example, for the E6 20% series there are six individual resistors or steps (1.0 to 6.8) and is given as the sixth root of ten ($\sqrt[6]{10}$).

Thus for the E12 10% series there are twelve individual resistor steps (1.0 to 8.2). Therefore it is given as the twelfth root of ten ($\sqrt[12]{10}$) and so on for the remaining E-series values.

TABLE 3. PREFERRED RESISTOR VALUES TABLE

Series	Preferred Resistor Values															
E-6	1.00		1.50		2.20		3.30		4.70		6.80					
E-12	1.00	1.20	1.50	1.80	2.20	2.70	3.30	3.90	4.70	5.60	6.80	8.20				
E-24	1.00	1.10	1.20	1.30	1.50	1.60	1.80	2.00	2.20	2.40	2.70	3.00	3.30	3.60	3.90	
	4.30	4.70	5.10	5.60	6.20	6.80	7.50	8.20	9.10							
E-96	1.00	1.02	1.05	1.07	1.10	1.13	1.15	1.18	1.21	1.24	1.27	1.30	1.33	1.37	1.40	
	1.43	1.47	1.50	1.54	1.58	1.62	1.65	1.69	1.74	1.78	1.82	1.87	1.91	1.96	2.00	
	2.05	2.10	2.15	2.21	2.26	2.32	2.37	2.43	2.49	2.55	2.61	2.67	2.74	2.80	2.87	
	2.94	3.01	3.09	3.16	3.24	3.32	3.40	3.48	3.57	3.65	3.74	3.83	3.92	4.02	4.12	
	4.22	4.32	4.42	4.53	4.64	4.75	4.87	4.99	5.11	5.23	5.36	5.49	5.62	5.76	5.90	
	6.04	6.19	6.34	6.49	6.65	6.81	6.98	7.15	7.32	7.50	7.68	7.87	8.06	8.25	8.45	
	8.66	8.87	9.09	9.31	9.53	9.76										

The above E-series values of Table 3, defines the different sets of standard resistor values. Thus E6 has six resistance values in each decimal interval, E12 has twelve, E24 has twenty-four and so on. Then each standard E-value is multiplied by 1, 10, 100, 1,000, 10,000, 100,000 or 1,000,000 in order to derive the desired nominal resistance value.

For example, the 1.20 value given in the E-24 series table will produce a series of resistors with resistances values of: 1.2 Ω , 12 Ω , 120 Ω , 1,200 Ω (or 1.2k Ω), 12k Ω , 120k Ω and 1.2M Ω each one with a 5% tolerance band.

The tolerance series of preferred values are ranges of resistor values chosen so that at maximum or minimum tolerance any one resistor overlaps with its neighbouring value. For example, take the E24 range of resistors with a 5% tolerance and resistor values are 47 and 51Ω respectively.

$47\Omega + 5\% = 49.35\Omega$, and $51\Omega - 5\% = 48.45\Omega$, an overlap of just 0.9Ω (49.35 – 48.45)

Note that all of the E-6 and E-12 values are also contained within the E-24 series but not all of them are contained within the E-96 Series table.

Five and six band colour coded resistors are more commonly associated with the high precision 1% and 2% film types while the 5% and 10% tolerances are for the more general purpose carbon types which tend to use the four band resistor colour code.

6. RESISTOR COLOUR CODES

A resistors value, tolerance, and wattage rating can be printed onto the body of the resistor as numbers or letters when the resistors body is big enough to read the print, such as large power resistors.

But when the resistor is small such as a 1/4 watt carbon or film type, these specifications must be shown in some other manner as the printed information would be too small to read, or over time and excessive heat, they could disappear completely.

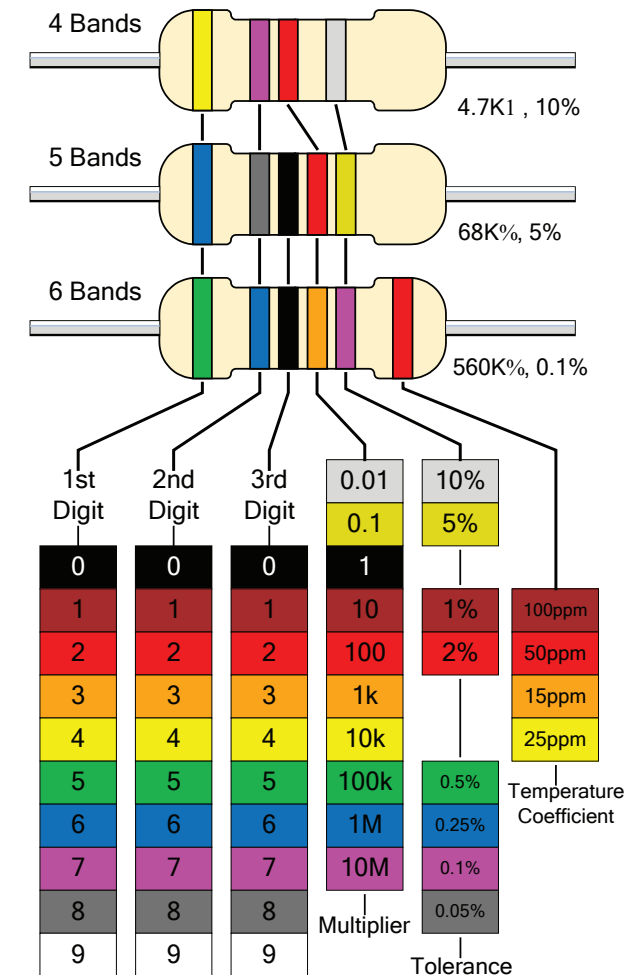
Resistor colour codes are a simple and visual way of identifying a resistors value and tolerance

To overcome this, manufacturers use a coloured banding system known commonly as a resistor colour code system of 4, 5 or 6 painted bands to indicate a resistors nominal value.

The body of small resistors use coloured painted bands to indicate both their resistive value and their tolerance with the physical size of the resistor generally indicating its wattage rating. These coloured painted bands can be used to easily identify a resistors ohmic value no matter what its size or condition without the need to measure it with the ohmmeter.

A resistors colour code markings consists of a set of individual coloured rings or bands in spectral order representing each digit of the resistors value. These coloured rings are always read one band at a time starting from the left to the right, with the larger width tolerance band oriented to the right side indicating its tolerance.

FIGURE 7. RESISTOR COLOUR CODES



By matching the colour of the first band with its associated number in the digit column of the colour chart below the first digit is identified and this represents the first digit of the resistance value.

The colour of the second band with its associated number in the digit column of the colour chart we get the second digit of the resistance value and so on. Then the resistor colour code is read from left to right as illustrated below in figure 7.

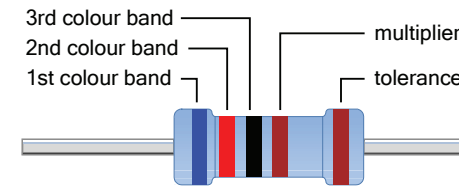
The “left-hand” or the most significant coloured band is the band which is nearest to a connecting lead with the colour coded bands being read from left-to-right as follows:

- **1st coloured band** – This coloured band is closest to the end of the resistor on the left hand side and represents the first digit of the resistors value.
- **2nd coloured band** – The second coloured band along representing the second digit of the resistors value.
- **3rd coloured band** – This is for metal film resistors which use a five or six-band colour code for more precise values.
- **4th coloured band** – This is the mathematical multiplier which represents the number of zeros to be added to the first two (or three) digits. If the third colour, for a 4-band resistor or the fourth colour, for a 5-band resistor, is either gold or silver, this represents a fractional decimal multiplier as the resistive value is less than 10Ω .

If the coloured band is Gold, multiply the first two or three digits by 0.1 (divide-by 10) and if the coloured band is Silver multiply by 0.01 (divide-by 100).

- **5th coloured band** – The final coloured band represents the tolerance of the resistor. A Gold band indicates a tolerance of $\pm 5\%$ while a Silver band indicates a tolerance of $\pm 10\%$. If there is no coloured band as in a three band resistor, then the tolerance is $\pm 20\%$.

FIGURE 8. FIVE-BAND RESISTOR COLOUR CODES



The coloured bands of the resistor given in figure 8 are shown as: **BLUE**, **RED**, **BLACK**, **BROWN** and **BROWN**.

The Blue coloured band is the first digit with a value of 6. The second Red band is the second digit value of 2. The third band is the third digit value of 0. This gives a three digit value of 620. The fourth band, Brown has a value of 10.

So the resistor has a resistive value of 6200 ohms ($620 \times 10 = 6200$) or $6k2\Omega$. The last band gives the resistors tolerance value and Brown equals a tolerance range of $\pm 1\%$. Thus the value of the example resistor is: **Blue Red Black Brown** = 6 2 0 0 = $6\ 2\ 0 \times 10 = \mathbf{6200\Omega}$ or **6k2 Ω $\pm 1\%$** .

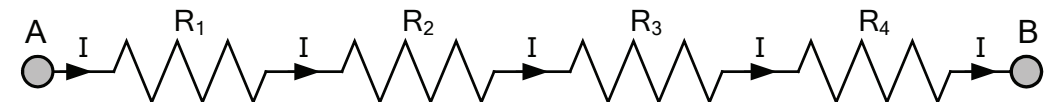
7. CONNECTING RESISTORS TOGETHER

If there is no one single resistor available for a specific resistance value, we can combine two or more resistors together to obtain the necessary value we require, either as a series or parallel combination. Individual resistors which are combined together within a circuit produce what is generally called an Equivalent Resistance value or R_{EQ} .

7.1 Resistors in Series

Individual resistors are said to be connected together in a “Series” combination when they are connected together in a single line as shown in Figure 9.

FIGURE 9. A SERIES CONNECTION



As all the current, (**I**) flowing through the first resistor of Figure 9 has no other path to flow through, it must therefore pass through the second resistor and then the third resistor and so on. Thus, resistors in series have a Common Current flowing through them because the current that flows through the resistor chain can only take one path.

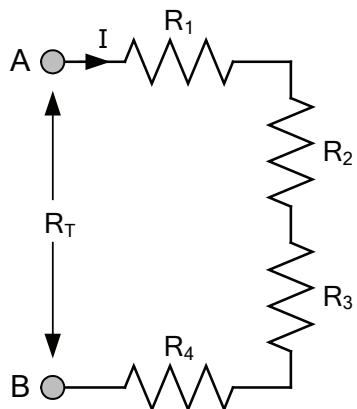
Thus for resistors in series: $I = I_{R1} = I_{R2} = I_{R3} = I_{R4} \dots \text{etc.}$

Since according to Ohms Law, the voltage drop across a resistor is the sum of the current times the resistance, $V = I \times R$, the voltage presented across the whole resistive combination is split between the individual resistors within the series chain.

Since the current through each resistor is the same, the voltage dropped across each resistor will depend only upon its resistance with the sum of all these voltage drops being equal to the supply voltage. Thus series resistive networks can also be thought of as "Voltage Dividers". So a series resistive circuit that has "N" resistors will have N-different voltage drops across it while maintaining a common current.

As the resistors are connected together in series the same current passes through each resistor in the chain, thus the total resistance, R_T between points A and B above must be equal to the sum of all the individual resistors in the series string added together. That is, R_T will equal $R_1 + R_2 + R_3 + R_4$.

FIGURE 10. A SERIES RESISTIVE CIRCUIT



Example: What will be the total resistance of the series circuit shown in Figure 10 if: $R_1 = 30\Omega$, $R_2 = 20\Omega$, $R_3 = 25\Omega$ and $R_4 = 45\Omega$

$$R_T = R_1 + R_2 + R_3 + R_4 = 30 + 20 + 25 + 45 = 120\Omega$$

Also, what will be the value of the current flowing around the circuit if 12 volts is applied between terminals A and B.

$$I = V \div R_T = 12 \div 120 = 0.1 \text{ Amps or 100 milliamperes}$$

What will be the voltage drops across each resistor?

$$V_1 = I \times R_1 = 0.1 \times 30\Omega = 3 \text{ volts}$$

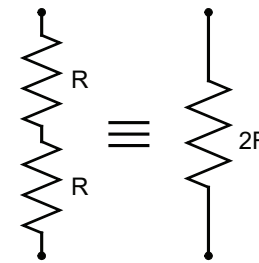
$$V_2 = I \times R_2 = 0.1 \times 20\Omega = 2 \text{ volts}$$

$$V_3 = I \times R_3 = 0.1 \times 25\Omega = 2.5 \text{ volts}$$

$$V_4 = I \times R_4 = 0.1 \times 45\Omega = 4.5 \text{ volts}$$

$$V_T = V_1 + V_2 + V_3 + V_4 = 3 + 2 + 2.5 + 4.5 = 12 \text{ volts}$$

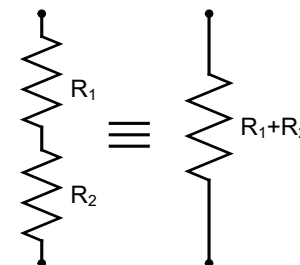
FIGURE 11 SAME RESISTIVE VALUE.



Since the total series resistance R_T of Figure 10. is equal to the sum of all the individual resistances added together, if two resistances in series string of Figure 11 are equal and of the same value, then the total or equivalent resistance, R_T will be equal to twice the value of one single resistor. That is equal to **2R**. For three equal resistors in series will be **3R**, etc.

Using resistors of the same value also means that the IR voltage drops across each individual resistance within the series string will be exactly the same so using Ohms Law above: $V_{R1} = V_{R2} = V_{R3} \dots \text{etc}$

FIGURE 12 DIFFERENT RESISTIVE VALUE.



If as shown in Figure 12, two resistors or impedances in series are unequal and of different values, then the total or equivalent resistance, R_T is equal to the mathematical sum of the two resistances.

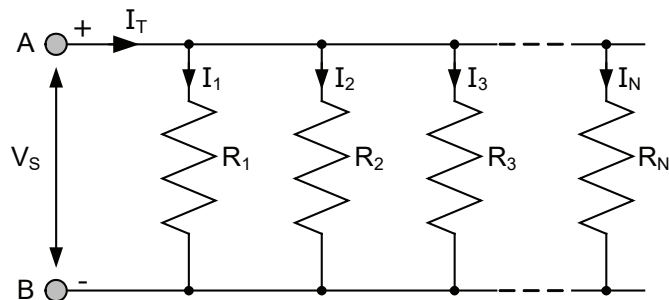
That is equal to $R_1 + R_2$. If three or more unequal (or equal) resistors are connected in series then the equivalent resistance is: $R_1 + R_2 + R_3 + \dots, \text{etc.}$

One important point to note about resistors in series networks and check that your maths is correct. The total resistance (R_T) of any two or more resistors connected together in series will always be GREATER than the value of the largest resistor in the series chain.

7.2 Resistors in Parallel

Individual resistors are said to be connected together in “Parallel” when both of their terminals are respectively connected to each terminal of the other resistor or resistors as shown in Figure 13. Unlike the previous series circuit, in a parallel resistor network the current can take more than one resistive path around the circuit.

FIGURE 13. A PARALLEL CONNECTION



Since there are multiple paths for the supply current to flow through, the current is not the same at all points in a parallel circuit. However, the voltage drop across all of the resistors in a parallel resistive network IS the same. Then, Resistors in Parallel have a Common Voltage across them and this is true for all parallel connected elements.

Thus for resistors in parallel: $V_S = V_{R1} = V_{R2} = V_{R3} = V_{RN} \dots \text{etc.}$

Since according to Ohms Law, the voltage drop across a resistor is the sum of the current times the resistance, $V = I \cdot R$. As the voltage across each resistor is the same, the current passing through each resistor will therefore depend on its resistive value, with the sum of all the parallel currents being equal to the supply current, I_T .

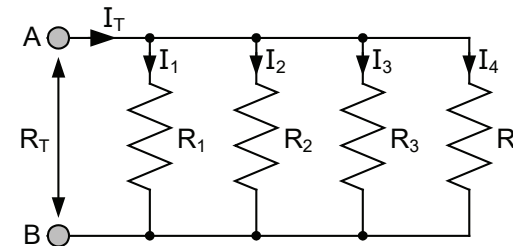
Thus parallel resistor networks can also be thought of as “Current Dividers” and a parallel resistor network having N resistors will have N-different current paths with the supply voltage presented across the whole parallel combination being the same.

Unlike the series resistor network whose total resistance, R_T is equal to the sum of all the individual resistors added together, the total or equivalent resistance for a parallel circuit is calculated differently.

Here, the reciprocal ($1/R$) value of the individual resistances are all added together instead of the resistances themselves with the inverse of the algebraic sum giving the equivalent resistance as shown.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \dots \dots + \frac{1}{R_N}$$

FIGURE 14 PARALLEL CONNECTED RESISTORS.



Example: What will be the total resistance of the circuit shown in Figure 14 if:

$R_1 = 10\Omega$, $R_2 = 25\Omega$, $R_3 = 60\Omega$ and $R_4 = 100\Omega$

$R_x = 1/R_1 + 1/R_2 + 1/R_3 + 1/R_4$

$R_x = 1/10 + 1/25 + 1/60 + 1/100 = 0.1667$

Therefore: $R_T = 1/R_x = 1/0.1667 = 6\Omega$

If a 12 volt battery is applied between terminals A and B, what will be the total current taken from the supply.

$$I = V_s \div R_T = 12V \div 6\Omega = 2.0 \text{ Amperes}$$

Then if 12 Amperes in total is delivered by the supply, what will be the individual branch currents flowing through each parallel resistor branch.

$$I_1 = V \div R_1 = 12 \div 10\Omega = 1.2 \text{ Amps}$$

$$I_2 = V \div R_2 = 12 \div 25\Omega = 0.48 \text{ Amps (480mA)}$$

$$I_3 = V \div R_3 = 12 \div 60\Omega = 0.2 \text{ Amps (200mA)}$$

$$I_4 = V \div R_4 = 12 \div 100\Omega = 0.12 \text{ Amps (120mA)}$$

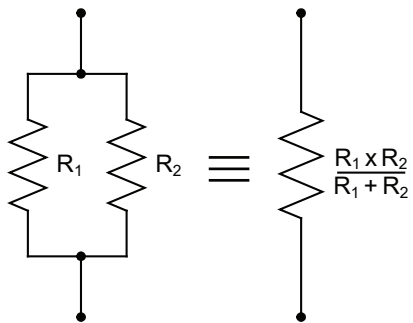
$$\text{Thus: } I_T = I_1 + I_2 + I_3 + I_4 = 1.2 + 0.48 + 0.2 + 0.12 = 2.0 \text{ Amps}$$

7.3 Two Resistors in Parallel

This method of reciprocal calculation can be used for calculating any number of individual resistances connected together within a single parallel network. If however, there are only two individual resistors in parallel like the ones shown in Figure 15, then a much simpler and quicker formula to find the total or equivalent resistance value, R_T and help reduce the reciprocal maths a little.

This much quicker product-over-sum method of calculating two resistors in a parallel combination, having either equal or unequal resistive values is given as:

FIGURE 15. ONLY TWO RESISTORS IN PARALLEL



$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

Thus, if $R_1 = 30\Omega$ and $R_2 = 45\Omega$, R_T will equal:

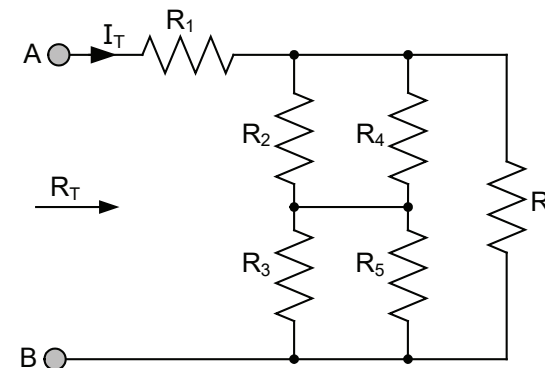
$$R_T = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{30 \times 45}{30 + 45} = 18 \Omega$$

An important point to remember about resistors in parallel. The total circuit resistance (R_T) of any two resistors connected together in parallel will always be LESS than the value of the smallest resistor in that combination.

7.4 Series-Parallel Resistive Networks

Resistive circuits generally combine series and parallel elements together to produce more complex resistive networks. Resistor circuits that combine series and parallel resistors together are generally known as Resistor Combination or mixed resistor circuits as shown in Figure 16.

FIGURE 16. RESISTOR COMBINATIONAL CIRCUIT



Resistors R_2 and R_4 are in parallel with each other. As are resistors R_3 and R_5

$$R_T = R_1 + (((R_2 \parallel R_4) + (R_3 \parallel R_5)) \parallel R_6)$$

$$I_T = V_{AB} \div R_T$$

Where the parallel bars “ $\parallel$ ” represents a parallel circuit condition.

The method of calculating a combinational circuits equivalent resistance, voltage drops, or branch currents is the same as that for any individual series or parallel circuit. We just need to remember that series circuits have a common current, (Voltage Division) while parallel circuits have a common voltage, (Current Division).

8. POWER IN RESISTORS

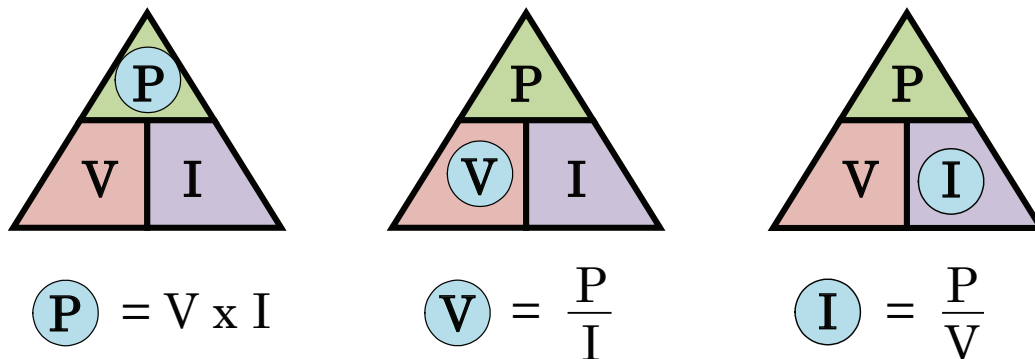
When an electrical current passes through a resistor due to the presence of a voltage across it, electrical energy is lost by the resistor in the form of heat and the greater this current flow the hotter the resistor will get.

Resistors are rated by the value of their nominal resistance as well as the electrical power given in watts, (W) that they can safely dissipate. A resistors power rating is sometimes called a *Resistors Wattage Rating*.

A resistors wattage rating is defined as the amount of heat that a resistive element can safely dissipate for an indefinite period of time without degrading its performance. Every resistor will have a maximum power rating and common values range from 1/4W up to 1000 Watts for ceramic or porcelain power resistors used for heating elements.

We know from above that when a current flows through a resistance, a voltage is dropped across it producing a product which relates to power. Thus if a resistance is subjected to a voltage, or if it conducts an electrical current, then it will always consume electrical power and we can superimpose these three quantities of power, voltage and current into a triangle called a Power Triangle as shown in Figure 17.

FIGURE 17. THE POWER TRIANGLE



Where: **P** is Power, **V** is Voltage and **I** is Current.

Resistor power rating is an important parameter to consider when choosing a resistor for a particular application. The electrical power dissipation of any resistor in a DC circuit can be calculated using one of the following three standard formulas:

$$\text{Power, } P = V \times I = I^2 R = \frac{V^2}{R}$$

Example: Calculate the power dissipated by a 240Ω resistor when connected to a 12 volt DC supply.

$$\text{Power, } P = V^2 \div R = 12^2 \div 240 = 0.6 \text{ watts}$$

As the power dissipated by a resistor is commonly linked to its physical size, a 1/4 (0.250) Watt resistor is therefore physically smaller than a 1 Watt resistor. Generally, resistors that are of the same ohmic value are also available in different power or wattage ratings. Carbon resistors, for example, are commonly made in wattage ratings of 1/8 (0.125)W, 1/4 (0.250)W, 1/2 (0.5)W, 1W, and 2 Watts, etc.

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With the completion of this Resistive Circuits eBook you should have gained a basic understanding and knowledge of resistors and how to connect them. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 5 we will learn about capacitors.

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